MILITARY SPECIFICATION

CONVERTER, DIGITAL-TO-DIGITAL CV-3166( )/ULR-17(V)

This specification is approved for use by the Electronics Research and Development Command (ERADCOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers the requirements for the construction and testing of the Converter, Digital-to-Digital CV-3166( )/ULR-17(V) (hereinafter referred to as Teletype Adapter A) which is part of the Control System of the AN/ULR-17(V) Receiving System (See Figure 1). Teletype Adapter A converts the digital command message created by the Control, Receiving Set C-9376 (hereinafter referred to as an RCU) into a teletype code suitable for use by standard military machines. The teletype code is such that, a print out by a teleprinter is in clear text.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bids or requests for proposal, form a part of the specification to the extent specified herein.

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, USAERADCOM, Fort Monmouth, NJ 07703, ATTN: DELET-DP, using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials-Handling, Wood
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade), and Cut Shapes
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Triple-Wall
PPP-S-760	Strapping, Nonmetallic (and Connectors)

MILITARY

MIL-P-116	Preservation, Method of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finish for Ground Signal Equipment
MIL-P-19644	Plastic Foam, Molded Polystyrene (Expanded Bead Type)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, For Packaging

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-147	Palletized and Containerized Unit Loads 40 inch x 48 inch Pallet
MIL-STD-188	Military Communications System Technical Standards
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment

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SPECIFICATIONS, MILITARY, STANDARDS (Cont'd)

MIL-STD-461	Electromagnetic Interference Characteristics,Measurement of
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1472	Human Engineering Design Criteria for Military Systems

DRAWINGS

USAERADCOM

DLSM-B-2014204	CV-3166( )/ULR-17(V) Data List
SM-D-2014204	CV-3166( )/ULR-17(V) Teletype Adapter A
SM-A-689132	Acceptance Test Procedure for Teletype Adapter A (A4)

(Copies of specifications, standards, and drawings required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The CV-3166 shall be constructed in accordance with the requirements of this specification and the drawings listed in the USAERADCOM Data List DLSM-B-2014204. The Teletype Adapter A CV-3166( )/ULR-17(V) top assembly drawing number is SM-D-2014204.

3.3 Parts, materials and process; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein shall apply.

3.3.1 Finish. The CV-3166 shall be finished in accordance with MIL-F-14072 and the drawings listed on DLSM-B-2014204.

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3.4 Marking.

3.4.1 General. Marking shall conform to MIL-M-13231 and MIL-STD-130 (See 4.4). Front panel marking shall be Group I as described in MIL-M-13231.

3.4.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.3 Serial numbers. Each CV-3166 shall have a serial number.

3.5 Performance.

3.5.1 Teletype adapter A CV-3166. The CV-3166 shall be capable of receiving control signals and the digital command message from the RCU C-9376 and convert portions of the command message to teletype code. The code shall be 10 bit ASCII or 7.0 baudot, as defined in MIL-STD-188, and selected by the insertion of the proper module into the CV-3166.

3.5.2 Control function. The CV-3166 shall be capable of creating a 21 character teletype message (See Table I) by constructing certain characters and extracting from the digital command work of the C-9376, information which is stored and converted into teletype characters. The CV-3166 shall transmit this message at either 60 wpm (words per minute) or 100 wpm teletype rate when the transmit switch is activated. The CV-3166 shall provide the function listed below.

FUNCTION

a. Preamble function.

b. Station address function.

c. Bandwidth function.

d. Frequency function.

e. Mode function.

f. Message identification function.

g. Message end function.

h. Unit power function.

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TABLE I - Character teletype message.

Character	Function
1	Machine function - "LETTERS"
2	Letter - "V"
3	Letter - "X"
4	Letter - "A"
5	Machine function - "FIGURES"
6	Station address - A number 0 thru 9
7	Bandwidth - A number 1 thru 9
8 thru 14	Frequency - 7 numbers 0 thru 9
15	Mode - A number 1 thru 5
16 thru 19	Message identification - 4 numbers 0 thru 9
20	Machine function - "CARRIAGE RETURN"
21	Machine function - "LINE FEED"

3.5.3 Preamble function. The preamble function is the first five characters of the message. These characters are constructed by the CV-3166 in the order of (1) the machine function "LETTERS", (2) letters "VXA", and (3) the machine function "FIGURES".

3.5.4 Station address function. The station address function is a ten position control, selectable from a front panel switch. With the selection of the address, the CV-3166 shall become associated with the CV-3167 which has been preset to the same address. An address of "0" shall address all CV-3167 unit(s) regardless of their switch setting.

3.5.5 Bandwidth function. The bandwidth character is extracted from the digital command word and then is converted into a number indicating bandwidth per Table II. The CV-3166 shall insert a number 7 in the bandwidth character when the retransmit switch is activated.

TABLE II - Bandwidth conversion.

PRINT NUMBER	HF BANDWIDTH	VHF BANDWIDTH
1	200 Hz	0.6 MHz
2	1.0 KHz	1.1 MHz
3	2.0 KHz	2.2 MHz
4	4.0 KHz	10 KHz
5	25.0 KHz	25 KHz
6	Spare	Spare
7	Request for Retransmit	Request for Retransmit
8	- - - - - - - - - - -	60 KHz
9	- - - - - - - - - - -	0.18 MHz

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3.5.6 Frequency function. The frequency function, which is seven digits, shall be extracted from the command message of the RCU C-9376 and converted into seven teletype characters. These seven characters will be numbers indicating the frequency to which the RCU C-9376 is tuned. The least significant character shall be 100 Hz. The seven frequency characters shall be indicated in the teletype message by characters 8 through 14.

3.5.7 Mode function. The mode function is extracted from the digital command message and converted into a teletype character. This character is a number indicating the mode per Table III and inserted into character 15 of the teletype message.

TABLE III - Mode conversion

PRINT NUMBER	MODE
1	AM
2	FM
3	CW/FSK
4	USB
5	LSB

3.5.8 Message identification.

3.5.8.1 Message identification function. The message identification function shall be four numbers representing the identity of the remote tuning message. They will be displayed on the front panel and inserted into characters 16 through 19 of the teletype message. The number shall be advanced one unit when the message is transmitted. It will not advance on retransmit.

3.5.8.2 Message identification setting. The setting of message identification numbers shall be controlled by four pushbutton switches. Each switch shall increment one digit of the message identification number each time it is depressed.

3.5.9 Message and function. The CV-3166 shall construct a machine function carriage return and line feed and insert these two characters into character 20 and 21 of the teletype message.

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3.5.10 Unit power function. The CV-3166 shall contain its own internal DC to DC power supply. The primary input power is +12 to ±0.6 Vdc. The power consumption of the CV-3166 shall be less than 15 watts.

3.5.11 Teletype message rate. The teletype output message rate is either 60 wpm or 100 wpm selectable by a rear panel switch.

3.5.12 Message activate. The output message of the CV-3166 shall be activated by a three position, spring loaded to return to center, toggle type switch. When placed in the down position it shall cause a complete control word to be transmitted. When placed in the up position it shall cause the transmission of the preamble and No. 7 bandwidth control as a request for retransmission.

3.6 Environmental. The CV-3166 herein after called equipment shall meet the following service conditions without degradation in specified performance of 3.5 unless specified herein.

3.6.1 Temperature altitude. The equipment shall be operable without degradation in specified performance at altitudes up to 30,000 ft above sea level and shall withstand air transportation at altitudes up to 50,000 ft above sea level. The equipment shall be capable of continuous operation without degradation in specified performance at ambient temperatures as low as -40 degrees F and as high as +131 degrees F, with intermittent operations to +160 degrees F. The equipment shall withstand exposure to ambient air temperature as low as -70 degrees F and as high as +160 degrees F. Testing shall be in conformance with the test requirements of 4.9.1.

3.6.2 Humidity. The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to relative humidities as high as 100 percent at all ambient air temperatures to 85 degrees F and relative humidities corresponding to a dew point of 85 degrees F at all temperatures between 85 degrees F and 120 degrees F, and low relative humidity of 5 percent at 120 degrees F.

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3.6.3 Sand and dust. The equipment shall withstand, in both operating and nonoperating conditions, exposure to fine sand and dust particles with wind speeds of up to 20 knots. After testing as specified in 4.9.3, the equipment shall meet the specified performance requirements.

3.6.4 Salt fog. In both operating and nonoperating conditions the equipment shall withstand the salt fog test of 4.9.4. Upon completion of test, the CV-3166, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surfaces; or exfoliation of metal. There shall be no clogging or binding of controls or other moving parts after subjecting the CV-3166 to the 48-hour salt fog exposure period or after subjection to the subsequent 48-hour drying period.

3.6.5 Fungus. After the fungus test of 4.9.5, the equipment shall be visually examined, using a 10-power magnifier. The CV-3166 shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.3.3) as evidenced by growth colonization (see 6.3.2) which includes branching (see 6.3.1) and sporulation (see 6.3.4) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.3.5) shall not be included in this evaluation.

3.6.6 Vibration and shock. The equipment shall withstand vibration and shock induced during field transport, by military vehicle, by rough handling, and mounted in military aircraft. Vibration isolation will be provided by the systems integrator when installed in the airframe.

3.6.7 Bench handling. The equipment shall withstand bench handling.

3.6.8 Explosive atmosphere. The equipment shall not cause ignition of an ambient-explosive gaseous mixture with air when operating in such an atmosphere.

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3.7 Electromagnetic interference (EMI). The equipment shall comply with the following emission and susceptibility requirements of Notice 4, MIL-STD-461A:

CE 01	CS 01	RE 02	RS 03
CE 03	CS 02		
CE 04	CS 06		
CE 05			

3.8 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (see 4.8).

3.9 Burn-in. Each unit on order shall meet the requirements herein, without subsequent processing, after subjection to burn-in as specified in 4.6.

3.10 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268.

General requirements for plastic material and parts
Wiring and cabling, Including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire cable

Flux and cleaning agent for soldering
Process for soldering
Cleaning
Riveting
General requirements for securing of parts
Cleaning after welding
Mechanical operation of controls
Process for welding
Gears
Self-tapping screws

b. In this specification:

3.3.1 Finish

3.11 Systems safety engineering.

3.11.1 Personnel hazards. Personnel hazards shall be kept to minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.10).

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3.11.2 Edge rounding exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.05 inch (1.3 mm).

3.11.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.12 Human factors engineering. Design configuration as affecting operation, maintainability and handling shall be maintained through compliance of configuration criteria to the requirements of MIL-STD-1472.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. First Article inspection (does not include preparation for delivery). (see 4.3).

b. Inspection covered by subsidiary documents. (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging. (see 4.5).

(2) Quality conformance inspection of packaging. (see 4.11).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

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4.3.1 First article units. The contractor shall furnish four first article units of Teletype Adapter A CV-3166( )/ ULR-17(V).

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified in Group A, Group B, and Group C (see Table V, VI and VII respectively). The inspection shall be performed in the following order: 4.4, Group A and Group B for all units and designated units for specific Group C tests in the order shown in Table IV.

TABLE IV - Environmental test for first article inspection.

INSPECTION (NOTE 1)	UNIT 1	UNIT 2	UNIT 3	UNIT 4
Vibration, equipment		1		
Temperature altitude	2			1
Shock, bench handling				
Humidity	3			
Fungus (Note 2)		2		
Salt fog (Note 2)			1	
Shock				2
Sand and dust (Note 2)				3
Explosive atmosphere			2	
EMI	1			

NOTE 1: Other first article inspection may precede, follow, or be interspersed between the tests.

NOTE 2: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification:

ITEM	WHERE REQUIRED
Finish Protective	3.3.1
Marking	3.4
Parts, Materials and Processes	3.3

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4.5 Quality conformance inspection of equipment before packaging. Each unit of each lot of equipment shall be burned-in prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table V. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the general inspection levels, and AQLs indicated in Table V.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

TABLE V - Group A inspections

INSPECTION	Req. Para.	Test Para.	AQL Major	AQL Minor
Visual and Mechanical	3.10	4.10	1.0%	4.0%
Interchangeability	3.8	4.8	1.0%	4.0%
Electrical				
Control Function	3.5.2	4.7.2	1.0%	*
Preamble Function	3.5.3	4.7.3	1.0%	*
Station Address	3.5.4	4.7.4	1.0%	*
Bandwidth Function	3.5.5	4.7.5	1.0%	*
Frequency Function	3.5.6	4.7.6	1.0%	*
Mode Function	3.5.7	4.7.7	1.0%	*
Message Identification Function	3.5.8.1	4.7.8	1.0%	*
Message Identification Setting	3.5.8.2	4.7.9	1.0%	*
Message End Function	3.5.9	4.7.10	1.0%	*
Unit Power Function	3.5.10	4.7.11	1.0%	*
Teletype Message Rate	3.5.11	4.7.12	1.0%	*
Message Activate	3.5.12	4.7.13	1.0%	*

*All electrical defects shall be considered major.

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4.5.2 Group B inspection. Group B inspection shall be formed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table VI and to the special inspection level of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans for the AQLs listed in Table VI shall be S-4. All defects shall be considered major.

TABLE VI - Group B inspection.

Inspection	Req. para.	Test para.	AQL
Interchangeability	3.8	4.8	6.5%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. Sample units shall be selected in accordance with 4.5.3.1. The inspection shall consist of the inspection specified in Table VII.

4.5.3.1 Sampling for Group C inspection. Four samples of the complete equipment shall be selected at random, 1 sample for each Subgroup I through and including Subgroup IV in Table VII. The first samples shall be selected at the start of production from the first quality conformance inspection lot. These samples shall constitute the Group C requirement for the first 25 units produced. Thereafter, 4 samples of the complete equipment shall be selected at random, 1 sample for each Subgroup I through Subgroup IV in Table VII from every 75 units and fraction thereof produced when the fraction is greater than or equal to one-half (1/2). When the fraction is less than one-half (1/2), the previous group upon which sampling was performed shall include the fractional quantity. Salt fog and fungus shall not be repeated after the first production lot Group C inspection.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order acceptable to the Government.

4.5.3.3 Group C failure. Action required relative to Group C failures shall be as specified in the contract or purchase order.

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4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are re-subjected to and pass Group A inspection.

TABLE VII - Group C inspection

Inspection	Req. para.	Test para.
Subgroup I		
Temperature/Altitude	3.6.1	4.9.1
Vibration	3.6.6	4.9.6
Subgroup II		
Humidity	3.6.2	4.9.2
Subgroup III		
Shock	3.6.7	4.9.7
Subgroup IV		
*Sand and Dust	3.6.3	4.9.3
*Salt Fog	3.6.4	4.9.4
*Fungus	3.6.5	4.9.5
Explosive Atmosphere	3.6.8	4.9.8
EMI	3.7	4.9.9

*The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.6 Burn-in. Every equipment shall be subjected to 100 hour burn-in. The last 24 hours of the burn-in shall be failure free. If an equipment fails during the last 24 hour failure free period, it shall be repaired, returned to test, and the test shall continue until a 24 hour failure free period has been achieved.

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4.6.1 Compliance. Units passing burn-in shall be submitted to the electrical measurements of Group A and B inspection without benefit of repair, cleaning, realignment or other changes, unless otherwise specified herein. If after successful completion of burn-in, the equipment fails either of the electrical measurements in Group A or Group B inspection, the equipment shall be reworked and resubmitted to the last 24 hours of the burn-in period and shall successfully complete the 24 hour failure free period of burn-in prior to being re-subjected to Groups A and B inspection.

4.6.2 Conditions preceding burn-in. A thermal survey and vibration survey shall be conducted in accordance with MIL-STD-781. The results of the required time to stabilize at the low temperature shall determine the length of the temperature cycle to be conducted during burn-in.

4.6.3 Test conditions. Each equipment shall be subjected to the following test conditions during burn-in:

Temperature -30 C to 55 C (-22 F to 131 F).

Temperature Cycling - Temperature cycle shall be timed to stabilize at low temperature as determined by the thermal survey, followed by timed to stabilize for six (6) hours at the high temperature.

Vibration 2.2G ± 10% peak acceleration value at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of operating time.

Equipment on-off cycling - Equipment off during the cooling cycle and on during the heating cycle.

Input Voltage - Nominal voltage of 12 Vdc ± 0.6V.

Input Voltage Cycling - The input voltage shall be maintained at 100% nominal for one-third of the equipment "on" cycle, at the nominal value for the second one-third of the equipment "on" cycle and at 90% for the final one-third of the equipment "on" cycle. This cycling procedure is to be repeated continuously throughout burn-in.

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4.6.4 Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from performance as required by this specification and the contractors approved test procedure.

4.7 Performance.

4.7.1 Teletype adapter A CV-3166. CV-3166 shall be tested in accordance with 5.1 of Test Procedure SM-A-689132 and shall meet the requirements of 3.5.1.

4.7.2 Control function. When the CV-3166 is tested in accordance with 5.1.1 through 5.1.8 of Test Procedure SM-A-689132, it shall meet the requirement of 3.5.2.

4.7.3 Preamble function. When tested in accordance with 5.1 of Test Procedure SM-A-689132 the converter shall meet the requirement of 3.5.3.

4.7.4 Station address function. The station address shall meet the requirements of 3.5.4 when tested in accordance with 5.1.9 of Test Procedure SM-A-689132.

4.7.5 Bandwidth function. The bandwidth function shall meet the requirements of 3.5.5 when tested in accordance with 5.1.9 of Test Procedure SM-A-689132.

4.7.6 Frequency function. The frequency function shall meet the requirements of 3.5.6 when tested in accordance with 5.1.9 of Test Procedure SM-A-689132.

4.7.7 Mode function. The mode function shall meet the requirements of 3.5.7 when tested in accordance with 5.1.9 of Test Procedure SM-A-689132.

4.7.8 Message identification function. The message identification function shall meet the requirements of 3.5.8.1 when tested in accordance with 5.1.9 of Test Procedure SM-A-689132.

4.7.9 Message identification setting. When the CV-3166 is tested in accordance with 5.1.3 of Test Procedure SM-A-689132, it shall meet the requirements of 3.5.8.2.

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4.7.10 Message end function. The message and function shall meet the requirements of 3.5.9 when tested in accordance with 5.1.9 of Test Procedure SM-A-689132.

4.7.11 Unit power function. When the CV-3166 is tested in accordance with 4.1 of Test Procedure SM-A-689132, it shall meet the requirements of 3.5.10.

4.7.12 Teletype message rate. The teletype message rate shall meet the requirements of 3.5.11 when tested in accordance with 5.1 of Test Procedure SM-A-689132.

4.7.13 Message activate. When the CV-3166 is tested in accordance with 5.1.8 and 5.1.10 of Test Procedure SM-A-689132, it shall meet the requirements of 3.5.12.

4.8 Interchangeability.

4.8.1 Inspection for dimensional interchangeability. The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirement of 3.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, keying pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major units.

e. Size and form of special threads.

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4.8.2 Electrical interchangeability. All like modules and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment as determined by the test specified in 3.8 or 4.8.1. However, electrical retuning and realignment of modules during production assembly operations may be expected and are permissible.

4.9 Environmental. The CV-3166 shall be tested in accordance with the methods and procedures of MIL-STD-810 given below. The limits of allowable performance degradation shall be as specified in 3.5 of this specification, and shall form the criteria for passing or failing each environmental test unless stated otherwise. Successful completion of any particular test shall be considered evidence of meeting the corresponding environmental condition.

4.9.1 Temperature-altitude. The equipment shall be subjected to the test of Method 504 of MIL-STD-810 for Class 4 equipment with the following changes and conditions:

a. The temperature for Step 1 shall be -70 degrees F.

b. The temperature for Steps 2 and 3 shall be -40 degrees F.

c. The procedure shall include a one hour exposure (nonoperating) to a pressure of 3.4 inches of Hg simulating air transportation of 50,000 feet at ambient temperature.

Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.5.

4.9.2 Humidity. The equipment shall be subjected to the test of Method 507, Procedure II of MIL-STD-810. In Step 6, measurements shall be taken during the 1st 5 hours of each cycle. Inability of the CV-3166 to meet the requirements of 3.6.3 shall constitute failure of this test.

4.9.3 Sand and dust. The equipment shall be tested in accordance with Method 510, Procedure I of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.5.

4.9.4 Salt fog. The equipment shall be tested in accordance with Method 509, of MIL-STD-810. Operation of the equipment shall not be required, and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the test items surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

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4.9.5 Fungus. The equipment shall be subjected to the fungus test specified in Method 508, Procedure I of MIL-STD-810. There shall be abundant growth colonization (6.3.2) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.6.5 shall constitute failure of this test.

4.9.6 Vibration. The equipment shall be subjected to the vibration test of MIL-STD-810, Method 514, Procedure I, Curve C, Part I, from Figure 514.2.2. Prior to testing, the equipment shall meet full specification performance for the measurements of 3.5.

4.9.7 Bench handling. The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in Table V.

4.9.8 Explosive atmosphere. The equipment shall be subjected to the test of Method 511, Procedure I of MIL-STD-810.

4.9.9 EMI. The equipment shall be tested for compliance with the requirements of 3.7. The test shall be in accordance with MIL-STD-462 as implemented by a contractor prepared, government approved EMI test plan.

4.10 Visual and mechanical inspection. The CV-3166 shall be examined for the defects listed in MIL-STD-252. An inspection shall also be performed to verify compliance with those portions of 3.11 (Systems Safety Engineering) which can be determined visually.

4.11 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of 5 herein. The rough handling test will be performed only when specified (see 6.2).

4.12 Human engineering verification. Maintenance of design configuration to the requirements of 3.12 shall be verified by visual and mechanical inspection.

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5. PACKAGING.

5.1 Preservation. Preservation shall be Level A, B or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Teletype Adapter A CV-3166( )/ULR-17(V) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Teletype Adapter A CV-3166( )/ULR-17(V) shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods/submethods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed submethod 1C-1.

5.1.1.4.2 Teletype adapter A CV-3166( )/ULR-17(V). Each Teletype Adapter A CV-3166( )/ULR-17(V) shall be individually unit packed method III as follows: Cushion on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane form conforming to MIL-P-26514. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be as specified in MIL-STD-1188.

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5.2 Packing. Packing shall be Level A, B or Commercial, as specified (see 6.2). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Teletype Adapter A CV-3166( )/ ULR-17(V), unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilities the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except the metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be laced on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.2 Level B. A quantity of Teletype Adapter A CV-3166( )/ ULR-17(V), unit packed as specified in 5.1, shall be packed within a close fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets constrapping conforming to QQ-S-781, type 1, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

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5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Teletype Adapter A CV-3166( )/ULR-17(V), unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I and II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The CV-3166 is one of high frequency components of Receiver Set, Countermeasure AN/ULR-17(V) which provides a monitoring capability of signals in the high frequency band of 0.5 to 30.0 MHz.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Marking and shipping of equipment.

c. Place of final inspection.

d. Technical literature required.

e. Government furnished property (loaned) for C-9376, CV-3167.

f. Level A, Level B or Commercial preservation and packing (see Section 5).

g. When rough handling test is required (see 4.11).

6.3 Definitions.

6.3.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (Hypha).

6.3.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.3.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.3.4 Sporulation. Sporulation is the formulation of minute unicellular reproductive or dormant bodies, called spores.

6.3.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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6.3.6 Exfoliation. Exfoliation is corrosion along with the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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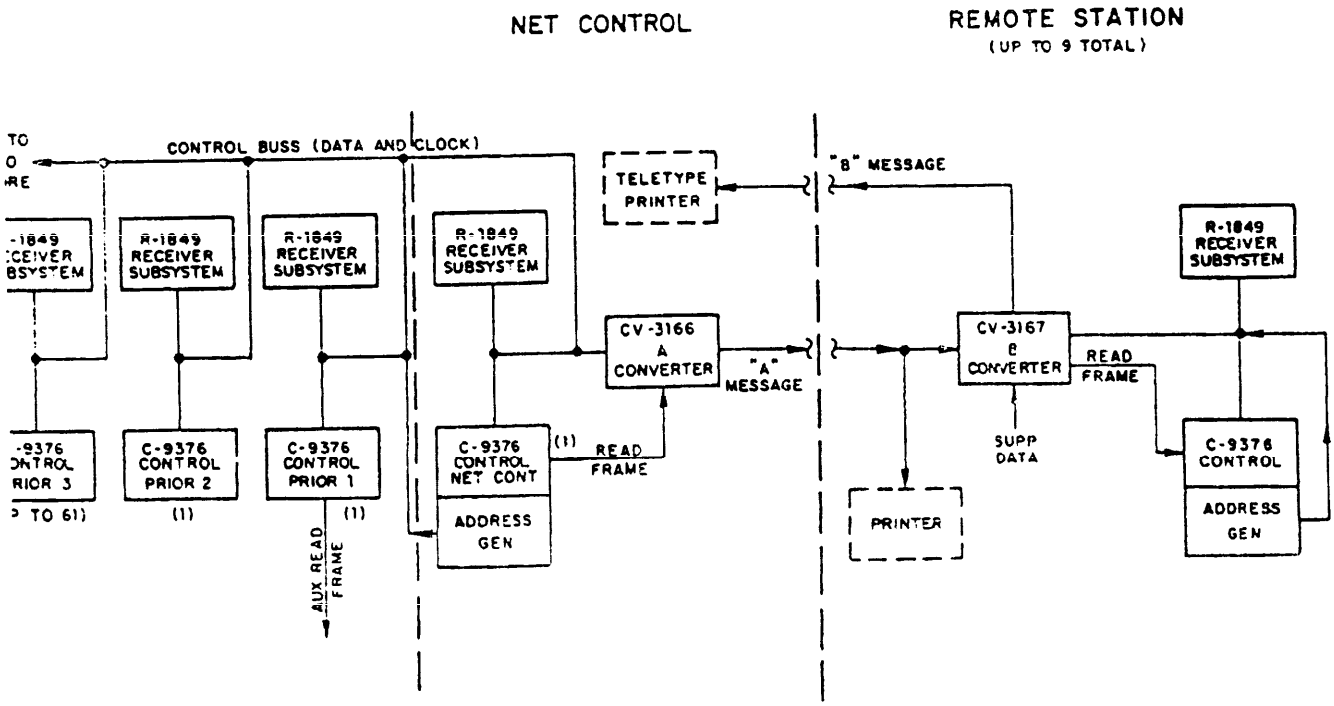


Figure 1- AN/ULR-17(V) (hf) system block diagram.

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DOCUMENT IDENTIFIER AND TITLE MIL-C-49351(ER)CONVERTER, DIGITAL-TO-DIGITAL CV-3166( )/ULR-17(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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